

Labwork 9: Histogram Equalization

Duong Tan Binh
Student ID: 2540007

1 Histogram

I use the grayscale image from Lab 4. Each thread reads one row and counts its gray levels in a separate 256-bin histogram. A second kernel adds the row histograms, with one thread per bin. Different threads never update the same row histogram, so atomic operations are not needed.

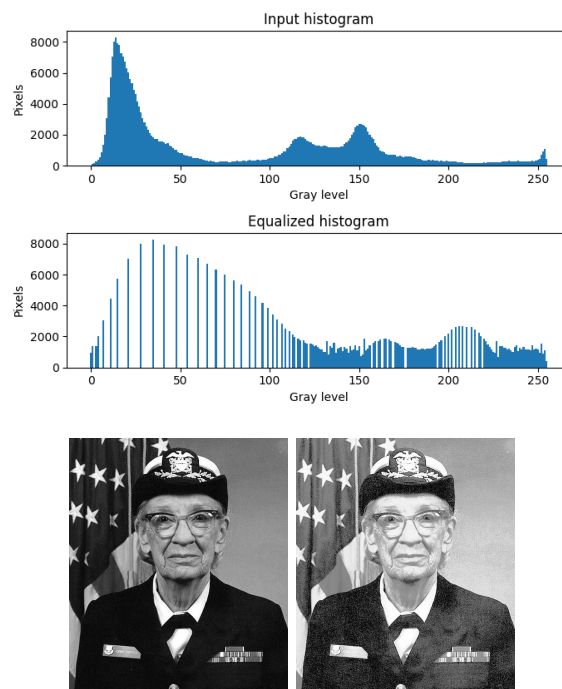
2 Equalization

I add the histogram counts in order on the CPU, as shown in the slide. For each level, I set its new value to $255 * \text{cumulative} / \text{pixelCount}$, keeping the integer part. This follows the CDF formula in the slide. The GPU then looks up the new value for every pixel.

3 Results

Pixels: 307200
Histogram total: 307200
Input range: 0 255
Output range: 0 255

The histogram counts and mapped pixels match the reference calculation. The output histogram is more spread out, but it is not exactly flat.



Input grayscale image and equalized result.